

Christina Chance

Community-Informed NLP · Algorithmic Fairness & Ethics
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EDUCATION

University of California at Los Angeles Ph.D. in Computer Science (M.S. awarded 2025) <i>Eugene V. Cota-Robles Fellow, GEM Fellow.</i> Advisor: Prof. Kai-Wei Chang	2022 – Present GPA: 3.85
Emory University B.S. in Computer Science and Mathematics, minor in Physics <i>Magna Cum Laude.</i>	2018 – 2022 GPA: 3.75

RESEARCH INTERESTS

My research focuses on (1) the application of NLP **methodologies for bias mitigation** and quantification in **social science**, (2) the study of **dialectal biases** in existing machine learning systems, and (3) the assessment and development of **community grounded content moderation approaches**. This culminates under the goal of engaging with, studying, assessing, and critiquing socio-technical systems that people engage with in their everyday lives as well as developing more inclusive and equitable systems.

RESEARCH EXPERIENCE

Adobe Research <i>Research Intern, advised by Jennifer Healey</i>	June 2025 – September 2025
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- ★ **Surprising Personas: Persona Sets with Controllably Unexpected Features [CHI EA 2026]**
 - Identified that demographic descriptor-based and feature-based synthetic persona sets produce preference gaps in LLM recommendation systems, where certain products are never recommended to specific demographic groups.
 - Introduced “surprising personas,” a controllable probabilistic generation method that increases recommendation diversity and entropy while better reflecting real user preferences, validated through a survey of 977 participants.

UCLA Natural Language Processing Lab <i>Graduate Student Researcher, advised by Prof. Kai-Wei Chang</i>	September 2022 – Present
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- ★ **IYKYK (But AI Doesn’t): Automated Content Moderation Does Not Capture Communities’ Heterogeneous Attitudes Towards Reclaimed Language [pre-print]**
 - Constructed and released an annotated corpus of reclaimed slur usage across Black, LGBTQIA+, and women communities, revealing that shared group identity does not produce reliable hate speech labeling agreement among in-group annotators.
 - Demonstrated that intra-community disagreement on reclaimed language interpretation reflects diverse lived experiences, highlighting the need for flexible, context-sensitive content moderation systems.
- ★ **Reinforcing Stereotypes of Anger: Emotion AI on African American Vernacular English [under-review, pre-print]**
 - Audited seven emotion AI models across 2.7 million geo-tagged LA tweets, finding false positive rates for anger on AAVE-dense text more than double those on General American English.
 - Linked neighborhood-level model predictions to Census demographics, revealing that anger predictions increase with African American population density, evidencing that emotion AI systems risk codifying racial stereotypes at scale.
- ★ **Research Domain Criteria and Deaths by Suicide in the National Violent Death Reporting System [JAMA Netw Open]**

- Applied LLMs to score RDoC psychopathology domains across 72,585 suicide death narratives in the National Violent Death Reporting System (NVDRS).
- Revealed that standard NVDRS mental health measures substantially undercount neurobehavioral dysfunction, with female and younger decedents carrying the highest burden.
- ★ **Re-imagining Virtual Communities: Ethical Guidelines for Studying Black Twitter [AIES 2025]**
 - Proposed an ethical checklist rooted in indigenous data sovereignty practices and ethnography to study digital marginalized communities.
 - Performed small qualitative and quantitative case study on Black Twitter users and the usage of the reclaimed n-slur.
- ★ **ModelCitizens: Representing Community Voices in Online Safety [EMNLP 2025]**
 - Created MODEL CITIZENS, a dataset of 6.8K posts and 40K annotations capturing diverse, community-informed toxicity perspectives with conversational context.
 - Fine-tuned LLAMACITIZEN-8B and GEMMACITIZEN-12B, outperforming GPT-4o-mini by 5.5% and exposing limits of current moderation tools.
- ★ **Measuring Narrative Complexity Among Suicide Deaths in the National Violent Death Reporting System (2003–2021 NVDRS) [MDPI Information 2025]**
 - Assess narrative complexity of the Suicide death narratives in the National Violent Death Reporting System leveraging NLP methods and approaches to capture syntactic complexity.
 - Highlight the need for more robust measures of complexity and quality of narratives as the records are used as tools to inform public health interventions.
- ★ **Will the Prince Get True Love’s Kiss? On the Model Sensitivity to Gender Perturbation over Fairytale Texts [NAACL TrustNLP 2025]**
 - Showed model sensitivity to gender perturbed fairytale text suggesting gender and stereotype bias present in the model finetuned on the FairyTaleQA dataset.
 - Proposed a novel method for Counterfactual Data Augmentation (CDA) using the power of Large Language Models (LLMs) to support a more robust domain of text.
- ★ **CORAAL QA: A Dataset and Framework for Open Domain Spontaneous Speech Question Answering from Long Audio Files [ICASSP 2024]**
 - Co-curated an extractive QA dataset from the Corpus of Regional African American Language (CORAAL), and designed a framework for spontaneous speech question-answering and information retrieval.
 - Leveraged LLMs to generate and evaluate additional training examples, assessing quality and usability for downstream tasks.
- ★ **An Analysis of Large Language Models for African American English Speaking Children’s Oral Language Assessment [NSBE Best Volume 1]**
 - Evaluated the performance of open-source automatic speech recognition systems (Whisper, HuBERT, Wav2Vec2) for the transcripts of primarily African-American English-speaking children.
 - Investigated a diverse set of metrics to assess AAE children’s speech for educational use to avoid biases around variations of the dialect compared to Standard American English.
- ★ **Leveraging Large Language Models and Topic Modeling for Toxicity Classification [ICNC 2025]**
 - Fine-tuned BERTweet and HateBERT on LDA-clustered toxic tweet datasets (3, 6, and 10 topics), achieving up to a 20% F1 improvement over GPT-4 and PerspectiveAPI.
 - Benchmarked models across 8 demographic subsets, reporting precision, recall, and F1 metrics while analyzing the impact of positionality bias on classification accuracy.
- ★ **Survey of Bias In Text-to-Image Generation: Definition, Evaluation, and Mitigation [pre-print]**
 - Conducted the first comprehensive survey of bias in text-to-image generative models, systematically reviewing 36 papers across three dimensions — gender presentation, skintone, and geo-cultural bias — and synthesizing their definitions, evaluation methods, and mitigation strategies
 - Identified key research gaps including the neglect of non-binary gender identities, lack of unified evaluation frameworks, and limitations of current mitigation approaches, outlining future directions toward human-centric bias research
- ★ **Leveraging Language Models to Detect Greenwashing [pre-print]**

- Developed a novel mathematical formulation to quantify greenwashing risk in corporate sustainability reports, fine-tuning ClimateBERT to achieve 86.34% accuracy and 0.67 F1 score on out-of-distribution test data from six real-world companies
- Designed a ground truth labeling pipeline combining sentiment, specificity, commitment, and hedging metrics, validated against expert annotations and real-world corporate carbon emissions data

Arnold Lab at Emory University

August 2021 – August 2022

Undergraduate Research Assistant/ Honors Thesis, advised by Prof. Dorian Arnold

★ **Zoom Audio Transcription Accuracy for African American Vernacular English [Emory Dis. Lib.]**

- Analyzed Zoom’s closed captioning accuracy for AAVE and SAE using conversational speech corpora, revealing mixed results and highlighting limitations of current tools for dialectal variation.

Bhasin Systems Biomedicine Lab at Emory University

August 2020 – August 2021

Undergraduate Research Assistant, advised by Dr. Manoj Bhasin

★ **Bioinformatics Tool Development & Database Engineering**

- Translated the TCRQuant command-line tool to a publicly accessible web server using PHP, JavaScript, and R, broadening access to t-cell receptor sequencing analysis for the research community.
- Engineered a gene splicing database capturing splice junction data for pediatric patients with Acute Myeloid Leukemia, supporting downstream clinical and genomic research.

WORK EXPERIENCE

Google LLC

May 2021 – August 2021

Software Engineering Intern, advised by Danny Qiu and Nolan Eastin

- Designed and implemented a feature to provide Google Cloud Storage customers with the ability to further protect their data by asymmetrically encrypting their Customer Supplied Encryption Key with a JSON payload using Java, Python, and C+.
- Utilized Tink, an open-source encryption library, and customer documentation to provide users with an approachable and improved encryption process.

Google LLC

May 2020 – August 2020

Student Training in Engineering Program Intern, advised by Sergei Karseko and Lian Wu

- Ideated, designed, and executed Covidiena, a web application for university students during the COVID pandemic to track health and stay updated with their school announcements and information.
- Implemented user authentication using HTTP Sessions and hashing to ensure individualized user experiences and allow multiple users using Cloud Datastore.

Google LLC

May 2019 – August 2019

Engineering Practicum Intern, advised by Filipe Ribiero and Tongfei Guo

- Created a system that generates simple explanatory summaries of top contributions to a change in metrics utilizing past data, differing dimensions, and relative metrics of interest with Python and GoogleSQL.
- Constructed backend that generated complex SQL statements based on performance metrics and manipulated data to perform trend analysis.

TEACHING EXPERIENCE

- **UCLA TA Coordinator** (Appointed Year-Long), 2025–2026
- **Teaching Assistant**, UCLA CS495 (TA Training Seminar), Fall 2025 — Prof. Sandra Batista. Supported ~75 students; graded assignments, provided feedback to new TAs, hosted office hours.
- **Ignite Fellow**, Teach for America, 2025.
- **Teaching Assistant**, UCLA CS131 (Programming Languages), Winter 2025 & Spring 2025 — Prof. Paul Eggert. Led weekly discussion sections and managed exams for ~120–180 students across two terms.
- **Teaching Assistant**, UCLA CS180 (Introduction to Algorithms and Complexity), Fall 2024 — Prof. Alexis Korb & Prof. Glenn Reinman. Supported ~240 students; led discussion sections and hosted office hours.

- **Teaching Assistant**, UCLA CS263 & CS162 (Natural Language Processing), Spring & Winter 2024 — Prof. Nanyun Peng. Helped create and grade assignments, exams, and course projects across ~120–130 students per term.
- **Teaching Assistant**, UCLA CS30 (Principles and Practices of Computing), Fall 2023 — Prof. Todd Millstein. Supported ~120 students.
- **Teaching Assistant**, Emory University CS377 (Database Systems), Spring 2022 — Prof. Nosayba El-Sayed. Supported ~40 students.

COMMUNITY ENGAGEMENT

- UCLA Student Conduct Member, 2024–Present.
- UCLA Student Fee Advisory Committee Member, 2024–Present.
- Computer Science Graduate Students Association at UCLA Executive Board Member, 2024–Present.
- Graduate Women in Computer Science at UCLA Treasurer & Executive Board Member, 2022–Present.
- Black Graduate Student Alliance at UCLA Historian & Treasurer, 2022–Present.
- Grad STRIVE 2.0 Grad2Grad Mentor, 2024–2025.
- ACM FAccT 2024 Craft Organizer of In Código of Hope: Possibilities of Global Solidarity and Activism in Academic Environments.
- UCLA Computer Science Summer Institute 2024 Introductory & Intermediate Program Coordinator.
- NAACL TrustNLP 2024 Program Committee.
- Institute for African-American Mentoring in Computing Sciences Mentor, 2023–2024.
- ACM SIGKDD KDD 2023 Equity, Diversity, Inclusion Day Workshop Co-Organizer and Panelist.
- ACM FAccT 2023 Volunteer.
- Black & Latinx in STEM at Emory University Co-Founder and President, 2019–22.

PROFESSIONAL AND ACADEMIC AFFILIATIONS

- Hartford Youth Scholars, Cohort 7.
- Ron Brown Scholars Captain, 2018.
- Girls for Technology, 2017–22.
- Initiative for Maximizing Student Development (IMSD) Scholar, 2020–22.
- Emory’s Scholarly Inquiry and Research at Emory (SIRE) Program, 2020–21.
- Alpha Sigma Chapter of Eta Sigma Pi, Spring 2020 Initiate.
- Emory University’s Circle of Omicron Delta Kappa, Spring 2020 Initiate.
- Nu Alpha Chapter of Alpha Kappa Alpha Sorority Incorporated, Spring 2021 Initiate.
- Emory University’s Chapter of Alpha Psi Omega, Spring 2022 Initiate.

AWARDS

- Outstanding Mentorship Award (Graduate Student), awarded by UCLA Computer Science 2026.
- GEM Fellow, awarded 2025. *Nationally competitive fellowship supporting underrepresented minority students in STEM PhD programs.*
- Eugene V. Cota-Robles Fellow, awarded by UCLA 2022–27. *Prestigious university-wide fellowship awarded to exceptional doctoral students from underrepresented backgrounds.*

- 100 Senior Honorary, awarded by Emory University 2022. *Recognizes the top 100 graduating seniors for academic and leadership excellence.*
- Computer Science Outstanding Student of the Year, awarded by Emory University 2022.

RELEVANT COURSEWORK

Undergraduate	Data Structures and Algorithms, Database Systems, Intro to Data Justice, Machine Learning Algorithms Artificial Intelligence, Analysis of Algorithms, Linear Algebra
Graduate	Deep Learning, Natural Language Processing, Mathematical Aspects of Scientific Computing: Linear Algebra, Fairness & Accountability in Natural Language Processing, Data Mining, Neuro-Symbolic Reasoning on Knowledge Graphs, Virtual Anthropology
Certification	AIM-AHEAD: NCATS Training Program - Cohort 2

SKILLS

Programming	Python (advanced) SQL, Java (proficient) R, PHP, MATLAB, HTML (basic)
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INTERESTS & ACTIVITIES

- **Dance:** Vibe Caribbean Dance Troop, performer and member, 2018–2022.
- **Theater:** Stage manager for two productions at Emory University, 2021–2022.
- **Athletics:** Track & Field, three-year varsity athlete, 2015–2018.